

What is the difference between __global__ and __host__ __device__?

Asked 1 year, 4 months ago Modified 1 year, 4 months ago Viewed 681 times



What is the difference between the following two declarations?

-1

```
__host__ __device__ void my_kernel(float* input, float* output, int size)
{
    // code to be executed by each thread in parallel
}
```



```
__global__ void my_kernel(float* input, float* output, int size)
{
    // code to be executed by each thread in parallel
}
```

As far as I understand, two copies of the same routine are created and executed in the first case.

How is the second case different from the first case?

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edited Mar 26, 2023 at 2:20



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asked Mar 25, 2023 at 20:51



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1 Answer

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The [programming guide defines them](#)

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__global__ is used to decorate a [kernel](#) definition in CUDA. A kernel is a function that will be auto-parallelized across multiple threads, as part of the kernel launch process. So a kernel launch requires a special handling like:



```
my_kernel<<<...>>>(...);
```



it would typically be "launched" like that in host code, but specifies execution on the device (host == CPU, device == GPU)



__host__ is used to decorate an ordinary C++ style function that is callable from host code (and runs on the host processor).

__device__ is used to decorate an ordinary C++ style function that is callable from device code and runs on the device processor (i.e. callable from a thread in a kernel, or another __device__ function.)

A function definition can have both __host__ and __device__ decorations, in which case a function will be compiled for the host, and another function of the same prototype will be compiled for the device. neither __host__ nor __device__ is used for decorating a kernel function, however.

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edited Mar 25, 2023 at 21:11

answered Mar 25, 2023 at 21:06



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